

VICKEY GHIMIRE

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Education

M.S. in Geospatial Information Science

The University of Texas at Dallas

08/2026 - 05/2028

Richardson, TX

B.S. in Computer Science

Northern Kentucky University | GPA: **3.94**/4.0

08/2022 - 05/2026

Highland Heights, KY

Technical Skills

Geospatial	GeoPandas, QGIS, rasterio, xarray, xrspatial, DEM Analysis, Multi-Criteria Evaluation, Spatial SQL, CRS/Projections, OSM, OSRM, Spatial Joins, Raster Processing
Programming	Python, C++, Java, SQL, JavaScript
Data & ML	Pandas, NumPy, PyTorch, SciPy, scikit-learn, Time-Series Modeling, NLP, LangChain
Web & Cloud	Django, FastAPI, PostgreSQL, REST APIs, AWS (ECS, RDS), Firebase

Experience

Full-Stack Django Developer

Department of Physics, NKU

09/2025 – 05/2026

Highland Heights, KY

- Engineered a modular, class-based Django platform with secure authentication, refactoring a legacy script into an MTV-compliant architecture, which reduced code-maintenance effort and enabled faster feature development.
- Built and integrated spatial data pipelines using OpenStreetMap and NASA POWER APIs to replace simulated inputs with site-level environmental data, improving accuracy of optimization workflows.
- Optimized a genetic-algorithm optimization workflow with convergence-based termination and result caching, cutting average computation time by **35%**.

Research Assistant- Data Analytics

Human-Computer Interaction Lab, NKU

10/2024 – 08/2026

Highland Heights, KY

- Designed an end-to-end novelty + relevance detection framework for LLM-generated content using hybrid semantic similarity (Jaccard + BERT), enabling automated content review of 9,000+ topics with **90%+** coverage.
- Led analysis of ChatGPT's memory feature via survey and privacy review of **135+** users; built reproducible pipelines to clean and transform multi-account datasets, automate chart generation, and streamline response analysis workflows.

Business Services Supervisor

NKU Campus Recreation Center

08/2025 – 05/2026

Highland Heights, KY

- Supervised and trained **25+** student employees across membership sales and operations at NKU's largest recreation facility; used membership analytics and reports to assist in staffing decisions and improve service quality.

Projects

Site Suitability Analysis: Fire Station in Kathmandu | *GeoPandas, xarray, rioxarray, rasterio*

July 2026

- Engineered a Multi-Criteria Evaluation (MCE) pipeline, aligning discrete OSM vectors and continuous SRTM DEM rasters into a unified raster grid for spatial overlay and suitability analysis.
- Computed slope gradients and Euclidean distance rasters to quantify terrain safety and emergency accessibility, normalizing results into classified 1–5 scores, flagging ideal locations to support operations and faster response times.
- Synthesized criteria via Weighted Linear Combination model, isolating a ranked shortlist of buildable candidate sites and exporting them as an interactive Leaflet map for ground evaluation.

Disaster-Aware Road Risk Viewer | *GeoPandas, React-Leaflet, OSRM, Shapely, FastAPI*

March 2026

- Developed a geospatial decision-support platform visualizing seasonal landslide/flood risk in Nepal's Bagmati Province using OSM road networks, DEM, and rainfall data.
- Segmented OSM networks into 1000m intervals, enriching road segments with terrain-derived slope and Terrain Ruggedness Index (TRI) to drive a multi-factor, non-linear risk engine.
- Implemented risk-aware routing by matching OSRM paths to pre-scored road segments, providing users with weighted average risk profiles and comparative safety analysis for travel paths.

Predictive Maintenance Framework | *Python, Scikit-learn, Pandas, MCP servers*

December 2025

- Built a two-stage predictive maintenance pipeline using Random Forest models to detect machine failures within 24-hours and estimate short-term Remaining Useful Life (RUL), achieving F1-score of **0.93** and R^2 of **0.98** for RUL prediction.
- Engineered temporal features from industrial telemetry and event logs (rolling stats, lag signals, maintenance histories) to capture pre-failure patterns that raw sensor data alone could not.